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Sri Lanka Institute of Information Technology

B.Sc. Honours in Computer Science /

B.Sc. Honours in Computer Systems Engineering

Final Examination
Year 2, Semester 1 (2024)

IE2004 – Computer Networks

Duration: 2 Hours

November 2024

Instructions to Candidates:

- This paper has 4 questions.
- Answer all questions in the booklet given.
- The total marks for the paper is 100.
- Electronic devices capable of storing and retrieving text and mobile phones are not allowed.
- Calculators are allowed.
- This paper contains 5 pages, including the cover page.

Question 1 (25 Marks)

a. With the aid of suitable diagrams, show how end stations are interconnected in a network with
   I. Bus topology (2 marks)
   II. Star topology (2 marks)

b. Determine the number of links in a network of six (6) nodes having full-mesh connectivity. (2 marks)

c. What are the three phases of data communication in a circuit switched network? (2 marks)

d. An end station A sends three packets to end station B via a packet switched network. What can you say about the paths traversed by the three packets if the network uses virtual circuit packet switching. (2 marks)

e. State the functions of each of the five layers of the TCP/IP protocol stack. Limit your answer to one sentence for each layer. (5 marks)

f. Figure 1 shows how a desktop computer A is connected to a server B. A sends an application layer message to B.
   I. Show how the application layer message is encapsulated in the link layer frame that leaves A, indicating the layer headers added by all layers. (4 marks)

   II. Describe which of the layer headers are processed by each of the following devices.
      i. Switch S1 (2 marks)
      ii. Router R (2 marks)

   III. Which of the layer headers in the frame leaving A remains unchanged all the way up to B? (2 marks)

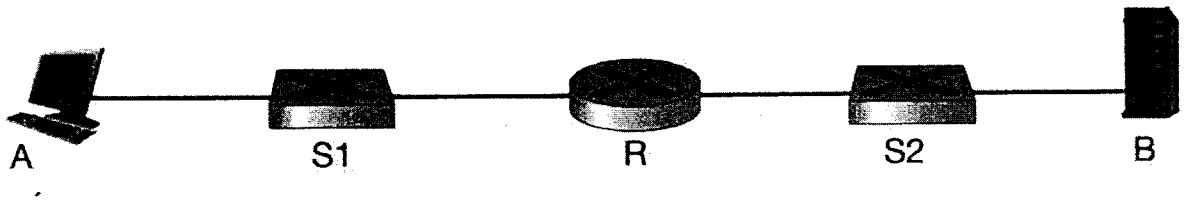


Figure 1

Question 2 (25 Marks)

a. Unshielded Twisted Pair (UTP) and Shielded Twisted Pair (STP) are two commonly used transmission media in local area networks.

I. State two features that are common to both STP and UTP. (2 marks)

II. Describe two different ways if shielding used in STP. (2 marks)

III. How is Near End Cross Talk (NEXT) affected due to shielding in STP? (3 marks)

b. State three benefits of using optical fiber media in networks. (3 marks)

c. Figure 2 shows a switched Ethernet network. The MAC address tables of all the three switches are initially empty. Two frames are sent in the network, in the following order.
Frame 1: From A to C
Frame 2: From C to A
Explain how the three switches S1, S2, and S3 forward each frame (selective forwarding or flooding) and show the MAC address tables of all three switches after each frame is forwarded. (8 marks)

d. VLANs need to be implemented in the network shown in Figure 2 by including hosts A and C in VLAN 100 and the other two hosts in VLAN 200. Determine the port type (access or trunk) of all switch ports, and the VLAN membership for each access port and the allowed VLANs for each trunk port. (4 marks)

e. Explain the difference between Passive scanning and Active scanning performed by a station before it joins an IEEE 802.11 wireless network. (3 marks)

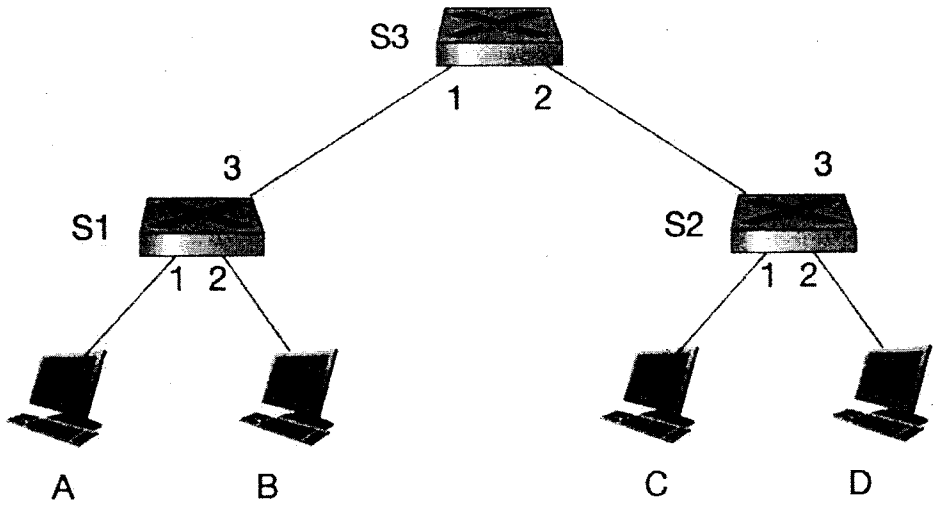


Figure 2

Question 3 (25 Marks)

a. The Time to Live (TTL) field is a part of the standard header of an IPv4 datagram.
   I. What does the TTL field indicate? (2 marks)
   II. How are routers forwarding the datagram required to change this field? (2 marks)
   III. What other field of the IPv4 datagram header requires to be changed due to the change in the TTL field? (2 marks)

b. State two differences between IPv4 and IPv6, which result in two of the fields in the IPv4 header being not included in the IPv6 header. (2 marks)

c. An institution requires IP addresses to be allocated for three networks of 500, 200 and 100 hosts respectively. The institution decides to use the private IPv4 address space 192.168.12.0/22.
   I. What are the address spaces allocated for the three networks? (4 marks)
   II. What are the lowest usable IPv4 address and the broadcast address in the network of 100 hosts? (2 marks)

d. Host A with IPv4 address 192.168.8.10 and MAC address aa:bb:cc:dd:ee:ff uses the ARP protocol to find the MAC address of host B which has the IPv4 address 192.168.8.20 and the MAC address ff:ee:dd:cc:bb:aa. Provide the Source and Destination MAC addresses of the Ethernet frame and the 4 address fields in the ARP message body of
   I. ARP request (3 marks)
   II. ARP response (3 marks)

e. The traceroute software tool uses a series of ICMP messages or UDP segments to determine the addresses of intermediate routers on the route to a particular destination.
   I. Briefly explain what field of the IPv4 header is used and how the value of that field is set in the series. (3 marks)
   II. What type of ICMP messages are received from the routers? (2 marks)

Question 4 (25 Marks)

a. A server application connected to port 100 on a host S is communicating with three clients connected to port 200 on host A, port 300 on host B, and port 400 on host C. Determine the number of sockets that would be used on host S, and give their endpoints (Source host and port, Destination host and port) if the application uses

I. UDP as the transport layer protocol (2 marks)

II. TCP as the transport layer protocol (2 marks)

b. A client host A is communicating with an echo server B listening on TCP port 2000. The client opens the TCP connection to the server, and then sends the lowercase text message "this is a test". When the message is received by the server, it changes the text of the message to uppercase and sends back to the client.

I. Show the flow of TCP segments between A and B in a timing diagram, including the flags (SYN, ACK) set in each segment. (5 marks)

II. The initial sequence numbers used by A and B are 10000 and 20000 respectively. For each of the segments listed below, determine the sequence number and the acknowledgement number in the TCP header, and the length (bytes) of data.

i. Segment used by A to send the text message to B (2 marks)

ii. Segment used by B to echo the modified text message to A (2 marks)

c. Briefly describe the exchange of DHCP messages between a client requesting an IPv4 address and a DHCP server. Indicate whether each message is sent as broadcast or unicast at the network layer. (4 marks)

d. A local DNS server is resolving the domain name "www.example.com" iteratively on behalf of a client. The DNS cache of the local DNS server is initially empty.

I. How many DNS queries is the server likely to make in the process? (1 mark)

II. For each query, specify the type of the server queried (root, TLD, authoritative) and the domain name for which the query is made. (3 marks)

e. Write Python code to implement the client described in Question 4(b). (4 marks)

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